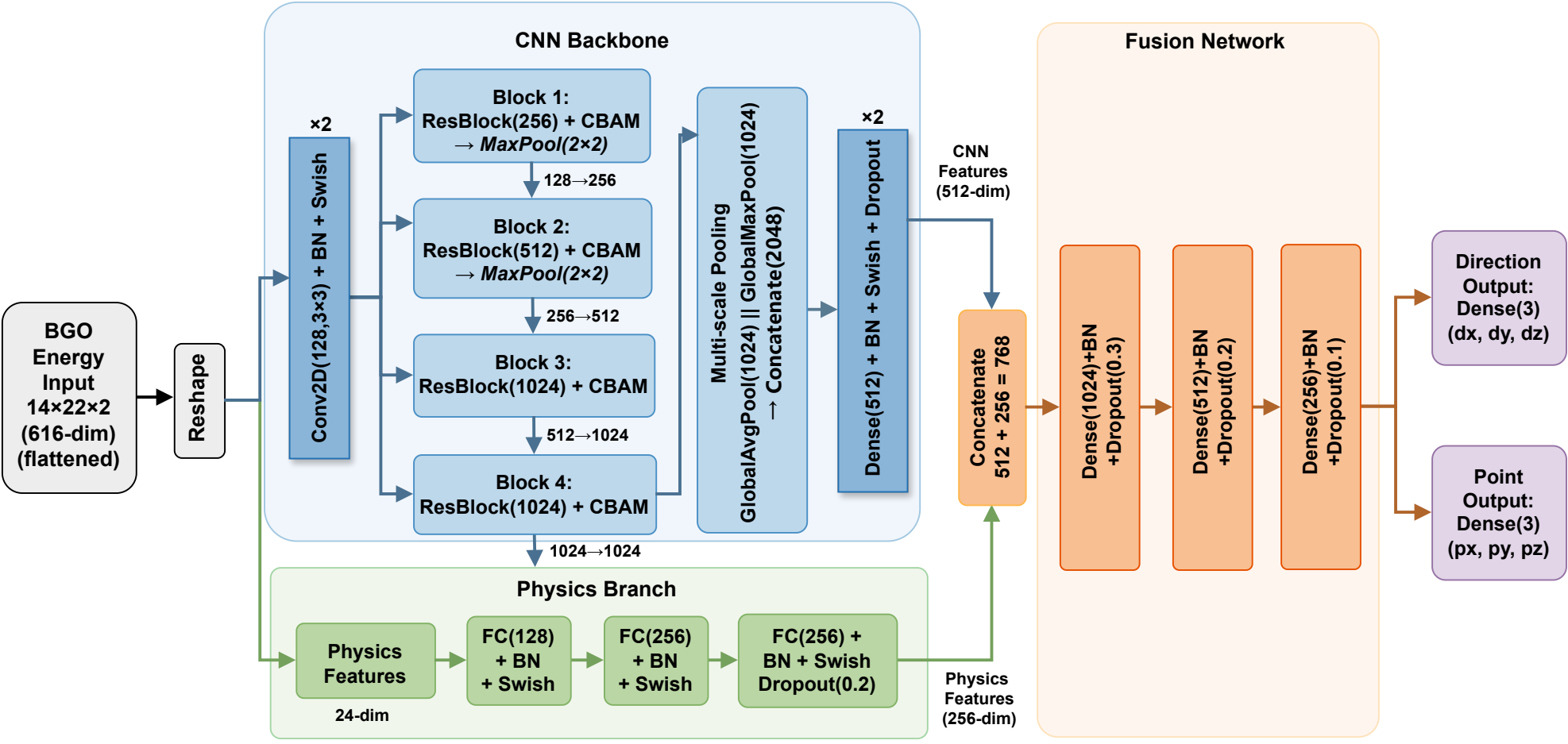

BGO Energy Input 14×22×2 (616-dim) (flattened)
Reshape
CNN Backbone
×2
Conv2D(128,3×3) + BN + Swish
Block 1: ResBlock(256) + CBAM → MaxPool(2×2)
128→256
Block 2: ResBlock(512) + CBAM → MaxPool(2×2)
256→512
Block 3: ResBlock(1024) + CBAM
512→1024
Block 4: ResBlock(1024) + CBAM
1024→1024
Multi-scale Pooling
GlobalAvgPool(1024) || GlobalMaxPool(1024)
→ Concatenate(2048)
×2
Dense(512) + BN + Swish + Dropout
Physics Branch
Physics Features
24-dim
FC(128) + BN + Swish
FC(256) + BN + Swish
FC(256) + BN + Swish Dropout(0.2)
Physics Features (256-dim)
CNN Features (512-dim)
Concatenate 512 + 256 = 768
Fusion Network
Dense(1024)+BN +Dropout(0.3)
Dense(512)+BN +Dropout(0.2)
Dense(256)+BN +Dropout(0.1)
Direction Output: Dense(3) (dx, dy, dz)
Point Output: Dense(3) (px, py, pz)